Lab 5: Exponential smoothing

MATH840 | Time Series | Week 6 — graded assignment 4 of 7

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What changes this week

Accuracy is marked from today. Your forecast is scored on the hidden horizon against your own benchmark, and that is 3 of the 8 points. Everything else works as in Week 4: the practice session is a worked example, the graded work is done at home, due **23:59 today**.

Open the notebook

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

The practice session's worked example, with the same helpers: Open in Colab

The task

Forecast **your series from Week 4** with exponential smoothing. The material is the Week 4 lecture, *Exponential Smoothing*.

The order matters, and it is marked:

1. **Specify the model by hand first** — error, trend, season, damping — from your own plots.
2. **Then** let `AutoETS` choose, and compare.
3. **Decide** between the candidates on a rolling origin, not on one window.
4. Check what is left in the residuals, and submit the forecast with the model's own 80% interval.

ETS only: no ARIMA, no regression, no averaging with other methods. Those come later, on this same series.

This one counts

8 points. **Deadline: 23:59 today**, on Moodle. The accuracy score is published the next morning; marks and comments come back before the next practice session.

What you are marked on

Section	Criteria	Points

Section	Criteria	Points
1	EDA and visualisation	1
2	Forecast accuracy against your benchmark	3
3	Justification block	3
4	Code quality and reproducibility	1
	Total	**8**

The notebook has one section per row. **Keep those headings.**

📊 Section 1 — EDA and visualisation (1 point)

You saw this series in Week 4. This time the plots have one job: to justify each letter of your specification **before** `AutoETS` runs.

- A time plot, read for **trend**: is there one, and does it flatten (damping)?
- The **seasonal swing against the level** — one point per year, max minus min of the year against its mean. A flat cloud says additive season; a cloud that climbs with the level says multiplicative. The template gives you `swing_against_level()`.
- The validation split on a plot: history, validation window, your chosen model's forecast and 80% interval over it.
- One sentence per plot. **A plot with no reading attached earns nothing.**

🎯 Section 2 — Forecast accuracy (3 points)

Scored from your submission file, on the `h` observations after your history that nobody has seen.

$$\text{skill} = \frac{\text{MASE of your forecast}}{\text{MASE of your benchmark}}$$

Skill	Points
below 1 — you beat your benchmark	3
1 to 1.1 — within 10%	2
above 1.1	1, and 3.3 must say why

- **Your benchmark** is the best of the four simple methods — `mean`, `naive`, `snaive`, `drift` — by RMSE on the last `h` observations of your history, refitted on all of it. The template computes it by exactly the scorer's rule, so the number you see is the one you are compared against.

- **MASE** is scaled by the in-sample seasonal naive error of your whole history — the same `mase()` as in the template.
- You are never compared with your classmates.

> ⓘ **What the pool guarantees, and what it does not**
>
> Every series in the pool was issued only if `AutoETS` **fitted on the whole history beats its benchmark** on the hidden horizon. That is a guarantee about `AutoETS` rerun on all of `y` — not about every ETS model, and not about the letters `AutoETS` picked on `train`.
>
> The Week 4 lecture's rule — shortlist on AICc, decide on a rolling origin — is still the right way to work, and it is what 3.2 pays for. Measured on the whole pool: shortlisting the three best models by AICc and choosing among them by a rolling origin beats the benchmark on **94%** of series and comes within 10% on 98%; choosing by a rolling origin among everything, 85%. If your evidence leads you away from `AutoETS`, follow it — and say in 3.2 what that evidence was.

✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

Item	Points	What earns them
3.1	1	**Your specification, before the machine.** Each letter of `MY_SPEC` and the damping choice, tied to a plot from Section 1. What in the plots would have made you choose differently.
3.2	1	**What `AutoETS` chose, and what you submitted.** Where `AutoETS` agrees or disagrees with you, and why. Your shortlist with AICc and rolling MASE. The model you submitted and the evidence for it — and if it is not `AutoETS`'s choice, what made you leave it.

Item	Points	What earns them
3.3	1	**What is left, and what you expect.** ACF and Ljung-Box on the **innovation** residuals, read: what structure is left and what it damages — the forecast or the interval. Your interval's coverage on the validation window. Your expected skill on the hidden horizon, and if you expect to lose, why.

Important

Marked on the reasoning. A specification by hand that `AutoETS` overrules, with a clear account of why, scores full marks in 3.1 and 3.2. A model that wins and cannot say why it was chosen does not.

Where this goes wrong

- **Specifying after the fact.** `MY_SPEC` copied from `AutoETS` and justified backwards reads exactly like that. Write 3.1 from Section 1 alone.
- **Choosing on one window.** The validation window is one draw of `h` observations. In the practice session it ranked a damped model first; over three windows the same model was last of the shortlist.
- **Comparing AICc across transformations.** AICc is comparable between ETS models on the same `y`; not between `y` and `log(y)`, and not against anything that is not ETS.
- `damped=None`. With `model="AAA"` and no `damped`, `AutoETS` may return `ETS(A,Ad,A)`. The template's `fit_ets()` passes `damped` explicitly — use it.
- **Testing the wrong residuals.** Ljung-Box goes on `fit.model_["residuals"]` of the chosen model, not on `y`.
- **Forecasting from `train`.** The submission is refitted on the **whole** history.
- **Freezing `AutoETS`'s letters from `train`.** If you choose `AutoETS`, submit `fit_ets(y)` — the search rerun on all of `y`. On half the pool it lands on different letters than it did on `train`, and the guarantee is about the rerun, not about the letters it found first.

📖 FPP3 ch. 8 · FPP 8.6 estimation and model selection · FPP 5.10 time series cross-validation · FPP the Pythonic Way, ch. 8

🖌 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: candidates are compared on `train` / `valid` and `rolling_mase`; only the final model is fitted on all of `y`.
- Every number in your text comes from a cell.

Submission

Upload three files to the Week 6 activity on Moodle **before 23:59 today**:

- `SURNAME_lab06_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
`student_id`	your identifier
`code`	your series code, the same as in Week 4
`ds`	the dates you are forecasting, in order
`yhat`	your point forecast
`yhat_lo_80`, `yhat_hi_80`	your model's 80% interval

- `SURNAME_lab06.pdf` — the report, with the justification block;
- `SURNAME_lab06.ipynb` — the source.

> ⚠ **A file that cannot be read scores 0 for accuracy**
>
> Exactly h rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine. The scorer reports format errors; it never repairs them.

How long this should take

Ninety minutes if you follow the practice session's example. `rolling_mase` refits every candidate three times, so a shortlist of six on a monthly series takes a minute or two to run.

For next week

Week 7 is Challenge 3, ARIMA, on this same series — the Week 6 lecture, *ARIMA Models, Part 1*, already published. Before then: run `nsdiffs` and `ndiffs` on your series, and look at the ACF and PACF of what is left after differencing. That is where Challenge 3's specification by hand starts.